

MATH 1342 – Elementary Statistics – Week 12: Inference for a Mean

Nov 9 – 11 | 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) • Veterans Day – No Class (Wed Nov 11) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain why inference for a mean uses the t distribution with $df = n - 1$ instead of the normal distribution. (CLO 5) 2. Check the conditions for t procedures ($n \geq 30$, or roughly normal data without extreme outliers). (CLO 5) 3. Construct and interpret a t confidence interval for a population mean, including the margin of error. (CLO 6) 4. Carry out a one-sample t test for μ and state the conclusion in context. (CLO 8)	e-Text: 6.2-CI Confidence Interval for a Mean e-Text: 6.2-D Distribution of a Mean e-Text: 6.2-HT Hypothesis Test for a Mean Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Mon Nov 16, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) In class Assignment Tool in Blackboard

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.

CLO 8 Perform hypothesis testing using statistical methods.